TTC from APS script

Tuesday, January 6, 2026 2:30 pm

- Time series of dimension T x P
  - T is number of time steps
  - P is the number of pixels
- I_p(t) is the intensity of the pth pixel at time t
- Elements defined by:
- $$M(t_1, t_2) = \sum_{p=1}^{P} I_p(t_1)\, I_p(t_2)$$
- And the per-frame normalisation is
- $$S(t) = \sum_{p=1}^{P} I_p(t), \qquad \alpha(t) = \frac{1}{S(t)}$$
- Normalised c2 then defined as:
- $$c_2(t_1, t_2) = P \cdot \frac{\sum_{p=1}^{P} I_p(t_1) I_p(t_2)}{\left(\sum_{p=1}^{P} I_p(t_1)\right)\left(\sum_{p=1}^{P} I_p(t_2)\right)}$$
- This is not the 'classical' XPCS
  - Unless you interpret the pixel-sum as the “ensemble average” over pixels in the ROI

Sanity check:

1. **Recompute TTC as G-TTC**
   - subtract mean
   - normalize by time-dependent σ
2. Compare:
   - Corr-TTC vs G-TTC
   - same mask, same data
3. Check whether:
   - oscillations persist
   - FFT peaks sharpen or disappear
   - diagonal artifacts vanish